

IR REMOTE CONTROL PROTOCOL

Arduino IRremote / Shirriff Library — Version 2.6.0

Protocol documentation for the Arduino robot project

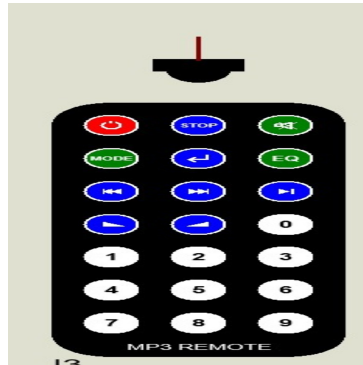


Figure 1. Actual MP3-style IR remote used with the robot.

1. Library and API Version

The robot firmware uses the classic 2.x API of the Arduino **IRremote** library associated with Ken Shirriff's original project. Version **2.6.0** is documented here. This is important because the supplied source uses the 2.x pattern **IRrecv**, **decode_results**, **irrecv.decode(&results;)**, **results.value**, and **irrecv.resume()**. Version 2.6.0 is a recognized 2.x release of the library. ■cite■turn0search0■turn0search11■

2. Receiving API Used by the Project

```
IRrecv irrecv(RECV_PIN);
decode_results results;
irrecv.enableIRIn();
if (irrecv.decode(&results;)) { ... }
irrecv.resume();
```

The 2.x API stores the decoded result in the **decode_results** structure, including **results.value**. The modern IRremote API changed this interface, so this documentation is specifically for the 2.6.0/2.x programming model.

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3. Protocol Identification

The command values used by this project match the common 32-bit NEC-style encoding represented by the legacy `IRremote results.value` field. For the documented commands, the values have the characteristic command/inverse-command structure:

Legacy value	Address / command interpretation
0xFF02FD	Address 0xFF, Command 0x02, inverse 0xFD
0xFF9867	Address 0xFF, Command 0x98, inverse 0x67
0xFFE01F	Address 0xFF, Command 0xE0, inverse 0x1F
0xFF906F	Address 0xFF, Command 0x90, inverse 0x6F
0xFFA857	Address 0xFF, Command 0xA8, inverse 0x57

The exact protocol interpretation is consistent with NEC-format command encoding, while the table above deliberately preserves the legacy hexadecimal values actually used by the project's Version 2.6.0 code.

4. Robot Command Mapping

Remote key / function	Legacy IR value	Robot action
Forward / Up	0xFF02FD	<code>forward()</code>
Backward / Down	0xFF9867	<code>backward()</code>
Left	0xFFE01F	<code>turnLeft()</code> , then stop
Right	0xFF906F	<code>turnRight()</code> , then stop
STOP	0xFFA857	<code>stopMotors()</code>
Repeat frame	0xFFFFFFFF	Ignored by command logic

5. Repeat-Frame Handling

The supplied firmware tests `results.value != 0xFFFFFFFF` before executing a command. This prevents the NEC repeat-frame value from being treated as a new movement command. After processing, `irrecv.resume()` re-enables reception of the next frame. The 2.x API documentation and examples use this same receive/decode/resume pattern. ■cite■turn0search1■turn0search7■

6. Implemented Decoding Sequence

- Receive an IR frame with `irrecv.decode(&results;)`.
- Check whether the received value is the repeat value `0xFFFFFFFF`.
- Print the decoded value in hexadecimal using `Serial.println(results.value, HEX)`.
- Compare the value with the programmed command constants.
- Execute the corresponding robot movement function.
- Store the value and call `irrecv.resume()`.

7. Example Project Code Pattern

```
if (irrecv.decode(&results;)) {  
  if (results.value != 0xFFFFFFFF) {  
    Serial.println(results.value, HEX);  
    if (results.value == 0xFF02FD) forward();  
    if (results.value == 0xFF9867) backward();  
    if (results.value == 0xFFE01F) { turnLeft(); delay(600); stopMotors(); delay(1000); }  
    if (results.value == 0xFF906F) { turnRight(); delay(600); stopMotors(); delay(1000); }  
    if (results.value == 0xFFA857) stopMotors();  
  }  
  key_value = results.value;  
  irrecv.resume();  
}
```

8. Technical Notes

Item	Documented value
Library family	IRremote / Shirriff 2.x
Version	2.6.0
Receive object	IRrecv
Result structure	decode_results
Decoded value field	results.value
Receiver restart	irrecv.resume()
Project protocol	NEC-style command encoding
Repeat value handled	0xFFFFFFFF

9. Compatibility Note

This document should be kept with the project if the repository is intended to reproduce the original firmware. Newer IRremote releases use a different API: `IrReceiver.decode()` and `IrReceiver.decodedIRData`. Therefore, changing the library version without adapting the code can cause compilation or behavior differences.

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10. Reference

The project should record **IRremote 2.6.0** in its README or dependency notes so that the original decoding interface and command values remain reproducible.